



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Metric in Function Spaces

Norms and Metrics in $C[a, b]$, $B(A)$, $L^p(E)$, and $L^\infty(E)$

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CONDUCTED BY

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Metric Space

Metric Space

A **metric space** is a pair (X, d) where $X \neq \emptyset$ and $d : X \times X \rightarrow \mathbb{R}$ satisfies, for all $x, y, z \in X$:

$$(M1) \quad d(x, y) \geq 0;$$

$$(M2) \quad d(x, y) = 0 \iff x = y;$$

$$(M3) \quad d(x, y) = d(y, x);$$

$$(M4) \quad d(x, z) \leq d(x, y) + d(y, z).$$

Here d is called a **metric** on X .

Normed Linear Space

Normed Linear Space

A **normed linear space** is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfies, for all $x, y \in X$ and $\alpha \in \mathbb{R}$ (or $\mathbb{C}$):

$$(N1) \quad \|x\| \geq 0;$$

$$(N2) \quad \|x\| = 0 \iff x = 0;$$

$$(N3) \quad \|\alpha x\| = |\alpha| \|x\|;$$

$$(N4) \quad \|x + y\| \leq \|x\| + \|y\|.$$

Here $\|\cdot\|$ is called a **norm** on X .

Metric Induced by a Norm

Let $(X, \|\cdot\|)$ be a normed space. Define $d : X \times X \rightarrow \mathbb{R}$ as

$$d(x, y) = \|x - y\|.$$

Show that d is a metric on X .

Function Spaces

Space of Continuous Functions $C[a, b]$

Let $a, b \in \mathbb{R}$ with $a < b$. The **function space** $C[a, b]$ is the set of all continuous functions defined on the closed interval $[a, b]$.

Space of Bounded Functions $B(A)$

Let A be a non-empty set. The **function space** $B(A)$ is the set of all bounded functions defined on A .

Norm and Metric in $C[a, b]$

💡 Max Norm in $C[a, b]$

Let $a, b \in \mathbb{R}$ with $a < b$. Define $\|\cdot\|_\infty : C[a, b] \rightarrow \mathbb{R}$ as

$$\|f\| = \max_{x \in [a, b]} |f(x)|.$$

Show that $\|\cdot\|$ is a norm on $C[a, b]$.



💡 Max Metric in $C[a, b]$

Let $a, b \in \mathbb{R}$ with $a < b$. Define $d : C[a, b] \times C[a, b] \rightarrow \mathbb{R}$ as

$$d(f, g) = \max_{x \in [a, b]} |f(x) - g(x)|.$$

Show that d_1 is a metric on $C[a, b]$.



💡 Integral Norm in $C[a, b]$

Let $a, b \in \mathbb{R}$ with $a < b$. Define $\|\cdot\| : C[a, b] \rightarrow \mathbb{R}$ as

$$\|f\| = \int_a^b |f(x)| dx.$$

Show that $\|\cdot\|$ is a norm on $C[a, b]$.



💡 Integral Metric in $C[a, b]$

Let $a, b \in \mathbb{R}$ with $a < b$. Define $d : C[a, b] \times C[a, b] \rightarrow \mathbb{R}$ as

$$d(f, g) = \int_a^b |f(x) - g(x)| dx.$$

Show that d is a metric on $C[a, b]$.



Norm and Metric in $B(A)$

💡 Supremum Norm in $B(A)$

Let A be a non-empty set. Define $\|\cdot\|_\infty : B(A) \rightarrow \mathbb{R}$ as

$$\|f\|_\infty = \sup_{x \in A} |f(x)|.$$

Show that $\|\cdot\|_\infty$ is a norm on $B(A)$.



💡 Supremum Metric in $B(A)$

Let A be a non-empty set. Define $d_\infty : B(A) \times B(A) \rightarrow \mathbb{R}$ as

$$d_\infty(f, g) = \sup_{x \in A} |f(x) - g(x)|.$$

Show that d_∞ is a metric on $B(A)$.



Problems with the Supremum and Integral Norm/Metric

Measure of a Set (Informal)

Lebesgue Integral (Informal)

What is L^p Space?

Formal Definition of $L^p(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. The **function space** $L^p(E)$ is the set of all equivalence classes of measurable functions $f : E \rightarrow \mathbb{R}$ such that

$$\int_E |f(x)|^p dx < \infty.$$

Here two functions f and g are said to be **equivalent** if $f(x) = g(x)$ for almost every $x \in E$.

Norm and Metric in $L^p(E)$

💡 L^p Norm in $L^p(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. Define $\|\cdot\|_p : L^p(E) \rightarrow \mathbb{R}$ as

$$\|f\|_p = \left(\int_E |f(x)|^p dx \right)^{1/p}.$$

Show that $\|\cdot\|_p$ is a norm on $L^p(E)$.

💡 L^p Metric in $L^p(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. Define $d_p : L^p(E) \times L^p(E) \rightarrow \mathbb{R}$ as

$$d_p(f, g) = \left(\int_E |f(x) - g(x)|^p dx \right)^{1/p}.$$

Show that d_p is a metric on $L^p(E)$.

The L^∞ Space

$L^\infty(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set. The **function space** $L^\infty(E)$ is the set of all equivalence classes of measurable functions $f : E \rightarrow \mathbb{R}$ such that

$$\operatorname{ess\,sup}_{x \in E} |f(x)| < \infty.$$

Here two functions f and g are said to be **equivalent** if $f(x) = g(x)$ for almost every $x \in E$.

💡 L^∞ Norm in $L^\infty(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set. Define $\|\cdot\|_\infty : L^\infty(E) \rightarrow \mathbb{R}$ as 

$$\|f\|_\infty = \operatorname{ess\,sup}_{x \in E} |f(x)|.$$

Show that $\|\cdot\|_\infty$ is a norm on $L^\infty(E)$.

💡 L^∞ Metric in $L^\infty(E)$

Let $E \subseteq \mathbb{R}$ be a measurable set. Define $d_\infty : L^\infty(E) \times L^\infty(E) \rightarrow \mathbb{R}$ as

$$d_\infty(f, g) = \operatorname{ess\,sup}_{x \in E} |f(x) - g(x)|.$$

Show that d_∞ is a metric on $L^\infty(E)$.

Thank You!

We'd love your questions and feedback.

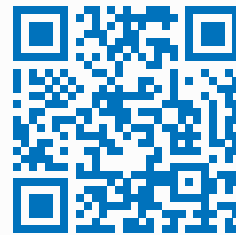
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.